

DSE511 Final Project Rubric

This rubric is designed for evaluating final projects in a PhD-level introductory data science course (e.g., DSE511). It will support the planned lightning talks (presentation) and final written reports, emphasizing scientific reasoning, reproducibility, and communication quality. Total: 100 points (30 for presentation, 70 for report).

Criterion	Weight	Exceeds Expectations (A, 9–10)	Meets Expectations (B, 7–8)	Approaches / Below Expectations (≤6)
Objective & Relevance	8 pts (3 pres + 5 report)	Problem/question clearly articulated, highly relevant to data science or student's domain, with explicit motivation and context.	Problem mostly clear; relevance stated but not fully justified or scoped.	Objective vague, misaligned, or missing.
Approach & Methods	16 pts (6 pres + 10 report)	Excellent methodological rationale and reproducibility; well- chosen tools, documented data sources, and transparent pipeline (e.g., Git/workflow).	Appropriate methods used with some justification and partial reproducibility.	Minimal or unclear methods; lacks reproducibility.
Results & Visualization	23 pts (8 pres + 15 report)	Clear, interpretable figures/tables; strong linkage between data and conclusions; insightful discussion of trends and limitations.	Results mostly clear; visualizations functional but not polished.	Limited or missing visualizations; little interpretation.
Schedule &	8 pts (3 pres	Evidence of planning,	Some planning	No evidence of

Execution	+ 5 report)	milestones met, and iteration/refinement; realistic and well-managed timeline.	evident; minor lapses in progress.	planning or progress tracking.
Future Directions & Reflection	15 pts (5 pres + 10 report)	Thoughtful next steps grounded in results; identifies limitations, ethical or reproducibility considerations, and broader impact.	Future ideas reasonable but brief or generic.	No reflection or forward-looking discussion.
Synthesis & Integration	5 pts (report only)	Integrates multiple topics coherently; demonstrates conceptual insight across course themes.	Some integration; mostly additive.	No attempt to connect ideas.
Communication Quality	15 pts (5 pres + 10 report)	Slide and report are professional, concise, well-designed, and free of major errors; excellent engagement and timing.	Clear and organized; minor issues in design or delivery.	Hard to follow or unprofessional.